

NE504 - Nuclear fuel cycle analysis

Two group diffusion

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1 Equation of continuity

The equation of continuity for neutrons describes behavior in a reactor.

[rate of change of neutrons] = [production rate] - [absorption rate] - [leakage rate]

$$\frac{d}{dt} \int_V n dV = \int_V s dV - \int_V \Sigma_A \phi dV - \int_V \underline{J} \cdot \underline{n} dA \quad (1)$$

The terms are -

- $\int_V n dV$ - total number of neutrons
- $\frac{d}{dt} \int_V n dV$ - rate of change
- $\int_V s dV$ - production rate
- $\int_V \Sigma_A \phi dV$ - absorption rate
- $\int_A \underline{J} \cdot \underline{n} dA$ leakage rate

Applying some mathematical relationships -

$$\begin{aligned} \frac{d}{dt} \int_V n dV &= \int_V \frac{\partial n}{\partial t} dV \\ \int_A \underline{J} \cdot \underline{n} dA &= \int_V \nabla \cdot \underline{J} dV \end{aligned} \quad (2)$$

Substitute into Eq. (1) -

$$\int_V \frac{\partial n}{\partial t} dV = \int_V s dV - \int_V \Sigma_A \phi dV - \int_V \nabla \cdot \underline{J} dV \quad (3)$$

Because the control volume is the same -

$$\frac{\partial n}{\partial t} = s - \Sigma_A \phi - \nabla \cdot \underline{J} \quad (4)$$

Apply Fick's law -

$$\begin{aligned} \frac{\partial n}{\partial t} &= D \nabla^2 \phi - \Sigma_A \phi + s \\ \frac{1}{v} \frac{\partial \phi}{\partial t} &= D \nabla^2 \phi - \Sigma_A \phi + s \end{aligned} \quad (5)$$

Assume steady state -

$$D \nabla^2 \phi - \Sigma_A \phi + s = 0 \quad (6)$$

2 Multigroup diffusion theory

The general multigroup diffusion equation based on Eq. (6) for an arbitrary group g is -

$$D_g \nabla^2 \phi_g - \Sigma_A^g \phi_g - \sum_{h=g+1}^N \Sigma_{g \rightarrow h} \phi_h + \sum_{h=1}^{g-1} \Sigma_{h \rightarrow g} \phi_h + s_g = 0 \quad (7)$$

In addition to absorption in group g , there is the loss due to scatter from group g to h but also a source terms due to scatter from h to g .

For 2 groups, then; i.e., a fast and thermal group, Eq. (7) is -

$$\begin{aligned} D_1 \nabla^2 \phi_1 - \Sigma_A^1 \phi_1 - \Sigma_{1 \rightarrow 2} \phi_1 + \Sigma_{2 \rightarrow 1} \phi_2 + s &= 0 \\ D_2 \nabla^2 \phi_2 - \Sigma_A^2 \phi_2 - \Sigma_{2 \rightarrow 1} \phi_2 + \Sigma_{1 \rightarrow 2} \phi_1 &= 0 \end{aligned} \quad (8)$$

Neutrons will not scatter from the thermal group (2) to the fast group (1) -

$$\begin{aligned} D_1 \nabla^2 \phi_1 - \Sigma_A^1 \phi_1 - \Sigma_{1 \rightarrow 2} \phi_1 + s &= 0 \\ D_2 \nabla^2 \phi_2 - \Sigma_A^2 \phi_2 + \Sigma_{1 \rightarrow 2} \phi_1 &= 0 \end{aligned} \quad (9)$$

3 Solution

1. Assume a one-dimensional, infinite slab.
2. Compute the derivatives -

$$\frac{d^2\phi_1}{dx^2} - \frac{1}{K^2}\phi_1 + d = 0 \quad (10)$$

$$\frac{d^2\phi_2}{dx^2} - \frac{1}{L^2}\phi_2 + \Sigma_{1 \rightarrow 2}\phi_1 = 0 \quad (11)$$

Where -

$$\begin{aligned} L^2 &= \frac{D_2}{\Sigma_A^2} \\ K^2 &= \frac{D_1}{\Sigma_A^1 + \Sigma_{1 \rightarrow 2}} \\ d &= \frac{s}{D_1} \end{aligned} \quad (12)$$

Let -

$$\frac{d^2\phi}{dx^2} \equiv \phi'' \quad (13)$$

3.1 Fast group

1. Assume the solution is the sum of a homogeneous and a particular solution.

$$\phi_1(x) = \phi_H(x) + \phi_P(x) \quad (14)$$

2. Making use of Eq. (10) -

$$\frac{d^2\phi_H}{dx^2} - \frac{1}{K^2}\phi_H = 0 \quad (15)$$

3. Assume the homogeneous solution -

$$Ae^{-\frac{x}{K}} + Ce^{\frac{x}{K}} \quad (16)$$

4. Assume the particular solution is an arbitrary constant -

$$\phi_P = B \quad (17)$$

5. Substitute Eq. (17) in to Eq. (10) -

$$0 - \frac{1}{K^2}B + d = 0 \quad (18)$$

6. The particular solution is then -

$$\phi_P = dK^2 \quad (19)$$

7. The fast flux is then -

$$\phi_1 = A_1e^{-\frac{x}{K}} + C_1e^{\frac{x}{K}} + dK^2 \quad (20)$$

3.2 Thermal flux

1. Substitute Eq. (20) into Eq. (11), using the shorthand notation (") for the second derivative -

$$\phi_2'' - \frac{1}{L^2}\phi_2 + \Sigma_1 A_1 e^{-\frac{x}{K}} + \Sigma_1 C_1 e^{\frac{x}{K}} + \Sigma_1 dK^2 = 0 \quad (21)$$

Where -

$$\Sigma_{1 \rightarrow 2} \equiv \Sigma_1 \quad (22)$$

2. Following the same procedure, assume the solution is the sum of a homogeneous and a particular solution.

$$\phi_2 = \phi_H + \phi_P \quad (23)$$

3. Making use of Eq. (11) -

$$\phi_H'' - \frac{1}{L^2}\phi_H = 0 \quad (24)$$

4. Assume the homogeneous solution -

$$\phi_H = A_2 e^{-\frac{x}{L}} + C_2 e^{\frac{x}{L}} \quad (25)$$

5. Apply the following [theorem](#) for the particular solution -

- (a) If $z_1(x)$ is a particular solution of -

$$y'' + p(x)y' + q(x)y = f(x) \quad (26)$$

- (b) and $z_2(x)$ is a particular solution of -

$$y'' + p(x)y' + q(x)y = g(x) \quad (27)$$

- (c) Then, for any number of forcing functions, $z = z_1(x) + z_2(x)$ is a particular solution of -

$$y'' + p(x)y' + q(x)y = f(x) + g(x) \quad (28)$$

6. Therefore, because there are three forcing functions in Eq. (21) -

$$\phi_P = \phi_F + \phi_G + \phi_I \quad (29)$$

7. Based on the theorem -

$$\begin{aligned} \phi_F'' - \frac{1}{L^2}\phi_F + \Sigma_1 A_1 e^{-\frac{x}{K}} \\ \phi_G'' - \frac{1}{L^2}\phi_G + \Sigma_1 C_1 e^{\frac{x}{K}} \\ \phi_I'' - \frac{1}{L^2}\phi_I + \Sigma_1 dK^2 \end{aligned} \quad (30)$$

The diffusion length (L^2) does not need to be split because the geometry is the same.

8. Assume the corresponding particular solutions for each forcing function -

$$\begin{aligned} \phi_F^P &= B_1 e^{-\frac{x}{K}} \\ \phi_G^P &= B_2 e^{\frac{x}{K}} \\ \phi_I^P &= B_3 \end{aligned} \quad (31)$$

9. Compute the second derivative of each particular solution -

$$\begin{aligned}\phi_F'' &= \frac{B_1}{K^2} e^{-\frac{x}{K}} \\ \phi_G'' &= \frac{B_2}{K^2} e^{\frac{x}{K}} \\ \phi_H'' &= 0\end{aligned}\tag{32}$$

10. Substitute Eq. (32) and Eq. (31) into Eq. (30) -

$$\begin{aligned}\frac{B_1}{K^2} e^{-\frac{x}{K}} - \frac{B_1}{L^2} e^{-\frac{x}{K}} + \Sigma_1 A_1 e^{-\frac{x}{K}} &= 0 \\ \frac{B_2}{K^2} e^{\frac{x}{K}} - \frac{B_2}{L^2} e^{\frac{x}{K}} + \Sigma_1 C_1 e^{\frac{x}{K}} &= 0 \\ 0 - \frac{B_3}{L^2} + \Sigma_1 d K^2 &= 0\end{aligned}\tag{33}$$

11. Simplifying -

$$\begin{aligned}\frac{B_1}{K^2} - \frac{B_1}{L^2} + \Sigma_1 A_1 &= 0 \\ \frac{B_2}{K^2} - \frac{B_2}{L^2} + \Sigma_1 C_1 &= 0 \\ \frac{B_3}{L^2} - \Sigma_1 d K^2 &= 0\end{aligned}\tag{34}$$

12. Solve for the constants -

$$\begin{aligned}B_1 &= \frac{\Sigma_1 A_1 K^2 L^2}{K^2 - L^2} \\ B_2 &= \frac{\Sigma_1 C_1 K^2 L^2}{K^2 - L^2} \\ B_3 &= \Sigma_1 d K^2 L^2\end{aligned}\tag{35}$$

13. Substitute the constants in Eq. (35) into Eq. (31) -

$$\begin{aligned}\phi_F^P &= \frac{\Sigma_1 A_1 K^2 L^2}{K^2 - L^2} e^{-\frac{x}{K}} \\ \phi_G^P &= \frac{\Sigma_1 C_1 K^2 L^2}{K^2 - L^2} e^{\frac{x}{K}} \\ \phi_I^P &= \Sigma_1 d K^2 L^2\end{aligned}\tag{36}$$

14. Finally, substitute the particular solutions in Eq. (36) and the homogeneous solution in Eq. (25) into Eq. (23) to obtain the thermal flux -

$$\phi_2 = A_2 e^{-\frac{x}{L}} + C_2 e^{\frac{x}{L}} + \frac{\Sigma_1 A_1 K^2 L^2}{K^2 - L^2} e^{-\frac{x}{K}} + \frac{\Sigma_1 C_1 K^2 L^2}{K^2 - L^2} e^{\frac{x}{K}} + \Sigma_1 d K^2 L^2\tag{37}$$

4 Solution check

4.1 Fast flux

1. The fast flux is -

$$\phi_1 = A_1 e^{-\frac{x}{K}} + C_1 e^{\frac{x}{K}} + dK^2 \quad (38)$$

2. And the solution must satisfy -

$$\frac{d^2 \phi_1}{dx^2} - \frac{1}{K^2} \phi_1 + d = 0 \quad (39)$$

3. Compute the second derivative of Eq. (38) -

$$\phi_1'' = \frac{A_1}{K^2} e^{-\frac{x}{K}} + \frac{C_1}{K^2} e^{\frac{x}{K}} \quad (40)$$

4. Substitute the result into Eq. (39) and simplify -

$$\left(\frac{A_1}{K^2} e^{-\frac{x}{K}} + \frac{C_1}{K^2} e^{\frac{x}{K}} \right) - \frac{1}{K^2} [A_1 e^{-\frac{x}{K}} + C_1 e^{\frac{x}{K}} + dK^2] + d = 0 \quad (41)$$

$$\frac{A_1}{K^2} e^{-\frac{x}{K}} + \frac{C_1}{K^2} e^{\frac{x}{K}} - \frac{A_1}{K^2} e^{-\frac{x}{K}} - \frac{C_1}{K^2} e^{\frac{x}{K}} - d + d = 0 \quad (42)$$

Check satisfied.

4.2 Thermal flux

1. The thermal flux is -

$$\phi_2 = A_2 e^{-\frac{x}{L}} + C_2 e^{\frac{x}{L}} + \frac{\Sigma_1 A_1 K^2 L^2}{K^2 - L^2} e^{-\frac{x}{K}} + \frac{\Sigma_1 C_1 K^2 L^2}{K^2 - L^2} e^{\frac{x}{K}} + \Sigma_1 dK^2 L^2 \quad (43)$$

2. And the solution must satisfy -

$$\frac{d^2 \phi_2}{dx^2} - \frac{1}{L^2} \phi_2 + \Sigma_1 \phi_1 = 0 \quad (44)$$

3. Compute the second derivative of Eq. (43) -

$$\phi_2'' = \frac{A_2}{L^2} e^{-\frac{x}{L}} + \frac{C_2}{L^2} e^{\frac{x}{L}} + \frac{\Sigma_1 A_1 K^2 L^2}{(K^2 - L^2) K^2} e^{-\frac{x}{K}} + \frac{\Sigma_1 C_1 K^2 L^2}{(K^2 - L^2) K^2} e^{\frac{x}{K}} \quad (45)$$

4. Substitute the result into Eq. (44) and simplify -

$$\begin{aligned} 0 = & \left[\frac{A_2}{L^2} e^{-\frac{x}{L}} + \frac{C_2}{L^2} e^{\frac{x}{L}} + \frac{\Sigma_1 A_1 K^2 L^2}{(K^2 - L^2) K^2} e^{-\frac{x}{K}} + \frac{\Sigma_1 C_1 K^2 L^2}{(K^2 - L^2) K^2} e^{\frac{x}{K}} \right] \\ & - \left[\frac{A_2}{L^2} e^{-\frac{x}{L}} + \frac{C_2}{L^2} e^{\frac{x}{L}} + \frac{\Sigma_1 A_1 K^2 L^2}{(K^2 - L^2) L^2} e^{-\frac{x}{K}} + \frac{\Sigma_1 C_1 K^2 L^2}{(K^2 - L^2) L^2} e^{\frac{x}{K}} + \Sigma_1 dK^2 \right] \\ & + [\Sigma_1 A_1 e^{-\frac{x}{K}} + \Sigma_1 C_1 e^{\frac{x}{K}} + \Sigma_1 dK^2] \end{aligned} \quad (46)$$

$$\begin{aligned}
0 = & + \left(\frac{A_2}{L^2} - \frac{A_2}{L^2} \right) e^{-\frac{x}{L}} + \left(\frac{C_2}{L^2} - \frac{C_2}{L^2} \right) e^{\frac{x}{L}} + (\Sigma_1 dK^2 - \Sigma_1 dK^2) \\
& + \left[\frac{\Sigma_1 A_1 K^2 L^2}{(K^2 - L^2) K^2} - \frac{\Sigma_1 A_1 K^2 L^2}{(K^2 - L^2) L^2} + \Sigma_1 A_1 \right] e^{-\frac{x}{K}} \\
& + \left[\frac{\Sigma_1 C_1 K^2 L^2}{(K^2 - L^2) K^2} - \frac{\Sigma_1 C_1 K^2 L^2}{(K^2 - L^2) L^2} + \Sigma_1 C_1 \right] e^{\frac{x}{K}}
\end{aligned} \tag{47}$$

$$\begin{aligned}
0 = & + \left(\frac{A_2}{L^2} - \frac{A_2}{L^2} \right) e^{-\frac{x}{L}} + \left(\frac{C_2}{L^2} - \frac{C_2}{L^2} \right) e^{\frac{x}{L}} + (\Sigma_1 dK^2 - \Sigma_1 dK^2) \\
& + \left[\frac{\Sigma_1 A_1 L^2}{K^2 - L^2} - \frac{\Sigma_1 A_1 K^2}{K^2 - L^2} + \frac{\Sigma_1 A_1 (K^2 - L^2)}{K^2 - L^2} \right] e^{-\frac{x}{K}} \\
& + \left[\frac{\Sigma_1 C_1 L^2}{K^2 - L^2} - \frac{\Sigma_1 C_1 K^2}{K^2 - L^2} + \frac{\Sigma_1 C_1 (K^2 - L^2)}{K^2 - L^2} \right] e^{\frac{x}{K}}
\end{aligned} \tag{48}$$

Check satisfied.